# Precalculus Review for Calculus

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Functions and composition**

Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 3$. Find:

**a** $(f \circ g)(x)$ $f(g(x)) = \sqrt{x^2+2}$.

**b** $(g \circ f)(x)$ $g(f(x)) = (x-1)+3 = x+2$, for $x \geq 1$.

**c** the domain of $f \circ g$ $x^2 + 2 \geq 1$ always holds, so the domain is all real numbers.

(a) $f(g(x)) = \sqrt{x^2+2}$.; (b) $g(f(x)) = (x-1)+3 = x+2$, for $x \geq 1$.; (c) $x^2 + 2 \geq 1$ always holds, so the domain is all real numbers.

**2** **[Foundational]** • **Target: Functions and composition**

Simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$ for $f(x) = x^2 - 3x$.

$\frac{(x+h)^2 - 3(x+h) - (x^2 - 3x)}{h} = \frac{2xh + h^2 - 3h}{h} = 2x + h - 3$.

**3** **[Foundational]** • **Target: Lines, exponentials, and trig — the toolkit calculus builds on**

Find the slope and equation of the line through $(-1, 4)$ and $(3, -2)$.

Slope $= \frac{-2-4}{3-(-1)} = -\frac{3}{2}$; equation $y - 4 = -\frac{3}{2}(x+1)$, i.e. $y = -\frac{3}{2}x + \frac{5}{2}$.

**4** **[Foundational]** • **Target: Lines, exponentials, and trig — the toolkit calculus builds on**

Solve $2e^{3x} = 14$ exactly and to three decimal places.

$e^{3x} = 7$, so $x = \frac{\ln 7}{3} \approx 0.649$.

**5** **[Foundational]** • **Target: Lines, exponentials, and trig — the toolkit calculus builds on**

State the exact value of each expression, needed frequently when differentiating trig functions:

**a** $\sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}$ **b** $\cos\frac{\pi}{3} = \frac{1}{2}$ **c** $\tan\frac{\pi}{6} = \frac{\sqrt{3}}{3}$

(a) $= \frac{\sqrt{2}}{2}$; (b) $= \frac{1}{2}$; (c) $= \frac{\sqrt{3}}{3}$

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**6** [Foundational] • Target: Factoring and rational expressions

Factor each expression fully:

**a** $x^2 - 5x + 6$ **b** $4x^2 - 9$ **c** $x^3 - 8$

$= (x-2)(x-3)$ $= (2x-3)(2x+3)$ $= (x-2)(x^2+2x+4)$

(a) $= (x-2)(x-3)$; (b) $= (2x-3)(2x+3)$; (c) $= (x-2)(x^2+2x+4)$

**7** [Foundational] • Target: Factoring and rational expressions

Simplify $\frac{x^2-4}{x^2-x-6}$, stating the value(s) of $x$ excluded from the domain.

$\frac{(x-2)(x+2)}{(x-3)(x+2)} = \frac{x-2}{x-3}$, excluding $x = -2$ (removable) and $x = 3$ (undefined).

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Inverse functions and logarithms

Find the inverse of $f(x) = 3x - 5$, and verify by computing $f(f^{-1}(x))$.

Swap and solve: $x = 3y - 5 \Rightarrow y = \frac{x+5}{3}$, so $f^{-1}(x) = \frac{x+5}{3}$. Check:
$f(f^{-1}(x)) = 3 \cdot \frac{x+5}{3} - 5 = x + 5 - 5 = x$. ✓

**9** [Intermediate] • Target: Piecewise and absolute value review

Sketch-reasoning: rewrite $f(x) = |2x - 6|$ as a piecewise function without absolute value bars.

$f(x) = 2x - 6$ for $x \geq 3$, and $f(x) = 6 - 2x$ for $x < 3$ (since $2x - 6 \geq 0$ exactly when $x \geq 3$).

**10** [Intermediate] • Target: Piecewise and absolute value review

Solve the inequality $|3x + 1| \leq 7$, writing your answer in interval notation.

$-7 \leq 3x + 1 \leq 7$: $-8 \leq 3x \leq 6$: $-\frac{8}{3} \leq x \leq 2$. Interval: $\left[-\frac{8}{3}, 2\right]$.

**11** [Intermediate] • Target: Systems of equations

Solve the system $2x + y = 7$ and $x - y = 2$ by elimination.

Adding the equations: $3x = 9$, so $x = 3$; then $y = 7 - 2(3) = 1$. Solution: $(3, 1)$.

**12** [Intermediate] • Target: Systems of equations

Find the point(s) of intersection of $y = x^2$ and $y = 2x + 3$.

$x^2 = 2x + 3$: $x^2 - 2x - 3 = 0$, $(x-3)(x+1) = 0$: $x = 3, -1$. Points: $(3, 9)$ and $(-1, 1)$.

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**13** [Intermediate] • Target: Graphing functions calculus will build on

The graph of $f(x) = x^3 - 3x$ is shown. Use it to estimate the $x$-intercepts and the local max/min locations — features calculus will let you find exactly.

From the graph, $x$-intercepts near $x = 0, \pm 1.7$; local max near $x = -1$, local min near $x = 1$ — these visual estimates match what derivative techniques will later confirm exactly.

**14** [Intermediate] • Target: Average rate of change (a preview of the derivative)

Find the average rate of change of $f(x) = x^2$ over $[1, 4]$, and interpret this as the slope of a secant line.

$\frac{f(4) - f(1)}{4 - 1} = \frac{16 - 1}{3} = 5$ — this is the slope of the secant line connecting $(1, 1)$ and $(4, 16)$.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Average rate of change (a preview of the derivative) • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

For $f(x) = x^2$, find the average rate of change over $[1, 1 + h]$ in terms of $h$, and describe what happens as $h \to 0$ (a preview of the derivative).

$\frac{f(1+h) - f(1)}{h} = \frac{(1+h)^2 - 1}{h} = \frac{2h + h^2}{h} = 2 + h$. As $h \to 0$, this approaches $2$ — the instantaneous rate of change (derivative) of $f$ at $x = 1$.

**16** [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 6.01 Related Rates

Simplify each expression, writing your answer with positive exponents:

**a** $x^{1/2} \cdot x^{1/3} = x^{5/6}$ **b** $(x^{2/3})^3 = x^2$ **c** $\frac{x^{5/2}}{x^{1/2}} = x^2$

(a) $= x^{5/6}$; (b) $= x^2$; (c) $= x^2$

**17** [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 6.02 Optimization and Curve Sketching

Rewrite $\sqrt[3]{x^2}$ using rational exponent notation, and evaluate it at $x = 8$.

$\sqrt[3]{x^2} = x^{2/3}$. At $x = 8$: $8^{2/3} = (8^{1/3})^2 = 2^2 = 4$.

**18** [Challenging] • Target: Interval notation and set operations review • Cross-Lesson: 7.01 Integration by Substitution

Write the solution set of $x^2 - 4 > 0$ in interval notation.

$x^2 > 4 \Rightarrow x < -2$ or $x > 2$. Interval: $(-\infty, -2) \cup (2, \infty)$.

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**19** [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 7.02 Integration by Parts

Level 1 · FoundationsFind the exact natural domain of the function $g(x) = \sqrt{\frac{x-1}{x+3}} - \ln(5-x)$ in interval notation.

We must evaluate two simultaneous restrictions:Square root radicand must be non-negative and denominator non-zero: $\frac{x-1}{x+3} \geq 0 \quad \text{and} \quad x + 3eq0.$ Sign analysis of the quotient with critical numbers $x = -3$ and $x = 1$: • For \( x Logarithm argument must be strictly positive: $x$

**20** [Challenging] • Target: Interval notation and set operations review • Cross-Lesson: 8.01 Area Between Curves

Level 3 · AdvancedLet $h(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. Prove that $h$ is strictly increasing on $\mathbb{R}$ without using calculus derivatives, deduce that $h$ is injective, and derive the explicit formula for $h^{-1}(x)$ including its domain.

Step 1 · Algebraic Monotonicity: Rewrite $h(x)$ by dividing numerator and denominator by $e^x$: $h(x) = \frac{1-e^{-2x}}{1+e^{-2x}} = \frac{(1+e^{-2x})-2e^{-2x}}{1+e^{-2x}} = 1 - \frac{2}{e^{2x}+1}$. As $x$, $\frac{2}{e^{2x}+1} \in (0,2) \Rightarrow h(x) = 1 - \frac{2}{e^{2x}+1} \in (-1,1)$. The domain of $h^{-1}$ is $(-1,1)$. Set $y = \frac{e^{2x}-1}{e^{2x}+1}$: $y(e^{2x}+1) = e^{2x} - 1 \Rightarrow ye^{2x} + y = e^{2x} - 1 \Rightarrow e^{2x}(1-y) = 1+y.$ For $y \in (-1,1)$ .

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Functions and composition

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown. The vertex of the parabola is at the point $(3, -5)$. Find the value of the parameter $k$.

**a.** $-5$

**b.** $4$ ✓

**c.** $9$

**d.** $0$

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. For the given function $f(x) = x^2 - 6x + k$, we have $a = 1$ and $b = -6$. Thus, the x-coordinate of the vertex is $x = -\frac{-6}{2(1)} = 3$. This matches the given vertex $(3, -5)$. Since the vertex $(3, -5)$ lies on the graph of the function, it must satisfy the function's equation. Substitute $x = 3$ and $f(x) = -5$ into the equation: $-5 = (3)^2 - 6(3) + k$ $-5 = 9 - 18 + k$ $-5 = -9 + k$ To solve for $k$, add 9 to both sides.

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**22** [Foundational] • Target: Lines, exponentials, and trig — the toolkit calculus builds on

Consider the graph of the function $f(x)$ shown below. Which of the following points lies on the graph of $f(x)$?

**a.** $(-2, 3)$

**b.** $(0, -1)$

**c.** $(1, 0)$ ✓

**d.** $(3, 4)$

The graph of $f(x)$ passes through the point $(1, 0)$. To verify this, we can observe the graph and identify the coordinates where the curve intersects the x-axis at $x = 1$. Let's check the other options. The point $(-2, 3)$ is not on the curve as the function value at $x = -2$ is approximately $5$. The point $(0, -1)$ is not on the curve as the function value at $x = 0$ is approximately $3$. The point $(3, 4)$ is not on the curve as the function value at $x = 3$ is approximately $0$. Therefore, only $(1, 0)$ lies on the.

**23** [Foundational] • Target: Factoring and rational expressions

Distance between skew lines $L_1 : \mathbf{r} = \mathbf{a} + s\mathbf{d}_1$ and $L_2 : \mathbf{r} = \mathbf{b} + t\mathbf{d}_2$:

**a.** $|\mathbf{b} - \mathbf{a}|$

**b.** $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$ ✓

**c.** $|\mathbf{d}_1 - \mathbf{d}_2|$

**d.** $0$

Applying Vectors and Vector-Valued Functions principles, Skew line distance formula. Therefore, the verified correct option is $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$.

**24** [Foundational] • Target: Inverse functions and logarithms

Let $F(x) = \int_2^x (t^2 - 1)dt$. According to the Fundamental Theorem of Calculus, what is the derivative of $F(x)$ with respect to $x$?

**a.** $x^2 - 1$ ✓

**b.** $x^2$

**c.** $x^2 - 1 - (2^2 - 1)$

**d.** $2x$

The Fundamental Theorem of Calculus, Part 1, states that if $F(x) = \int_a^x f(t)dt$, then $F'(x) = f(x)$. In this problem, $f(t) = t^2 - 1$ and the lower limit of integration is $a = 2$. Therefore, $F'(x) = x^2 - 1$.

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**25** [Foundational] • Target: Piecewise and absolute value review

What is the maximum rate of change of $f(x, y) = x\cos(y) - 3y$ at $(2,\ 3)$?

**a.** $3\sin^2(3) + 12\sin(3) + 10$

**b.** $2\sin(3) - \cos(3) + 3$

**c.** $\sqrt{3\sin^2(3) + 12\sin(3) + 10}$ ✓

**d.** $2\sin(3) + 3$

The maximum rate is $||ablaf||$. Here $ablaf(2,\ 3) = \langle\cos(3),\ -3 - 2\sin(3)\rangle$, so it is $\sqrt{3\sin^2(3) + 12\sin(3) + 10}$.

**26** [Foundational] • Target: Systems of equations

Consider the function $f(x) = 2x^2 - 3x + 1$. What is the value of $f(3)$?

**a.** 10

**b.** **13** ✓

**c.** 16

**d.** 19

To find the value of $f(3)$, we substitute $x = 3$ into the function $f(x) = 2x^2 - 3x + 1$. $f(3) = 2(3)^2 - 3(3) + 1$ $f(3) = 2(9) - 9 + 1$ $f(3) = 18 - 9 + 1$ $f(3) = 9 + 1$ $f(3) = 10$. Therefore, $f(3) = 10$.

**27** [Foundational] • Target: Graphing functions calculus will build on

A spherical balloon is inflated at 4 cm³/s. Rate of radius change when $r = 2$:

**a.** $\frac{1}{\pi}$

**b.** $\frac{1}{2\pi}$

**c.** $\frac{1}{4\pi}$ ✓

**d.** $\frac{1}{8\pi}$

From $V = \frac{4}{3}\pi r^3$, we have $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 2$ and $\frac{dV}{dt} = 4$, $\frac{dr}{dt} = \frac{4}{4\pi(4)} = \frac{1}{4\pi}$ cm/s.

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**28** [Foundational] • Target: Average rate of change (a preview of the derivative)

Symmetric equations of line through $(x_0, y_0, z_0)$ with direction $(a, b, c)$:

**a.** $ax + by + cz = 0$

**b.** $x/a = y/b = z/c$

**c.** $\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$ ✓

**d.** $x + y + z = 1$

Applying Vectors and Vector-Valued Functions principles, Symmetric form. Therefore, the verified correct option is $\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$.

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Radicals and rational exponents

Net change in population from $t = 0$ to $t = T$ with rate $r(t)$:

**a.** $r(T) - r(0)$

**b.** $\int_0^T r(t)\,dt$ ✓

**c.** $r'(T)$

**d.** $r(0) \cdot T$

Applying Applications of Integration principles, Net change = integral of rate. Therefore, the verified correct option is $\int_0^T r(t)\,dt$.

**30** [Intermediate] • Target: Interval notation and set operations review

A 200-L tank holds brine with 20 kg of salt. Brine with $\frac{1}{2}$ kg/L enters at 10 L/min and the well-mixed solution leaves at the same rate. How much salt (kg) is in the tank after 20 min?

**a.** $100 - \frac{80}{e}$ ✓

**b.** $100 - \frac{100}{e}$

**c.** $120$

**d.** $\frac{20}{e} + 100$

$A' = 5 - \frac{10}{200}A$, a linear equation with equilibrium $100$. So $A = 100 - 80e^{-\frac{t}{20}}$ and $A(20) = 100 - \frac{80}{e}$.

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**31** [Intermediate] • Target: Functions and composition

Directional derivative of $f = \sin(xy)$ at $(0, \pi)$ in direction $\hat{u} = (1/\sqrt{2}, 1/\sqrt{2})$:

**a.** $\pi$

**b.** $0$

**c.** $-\pi/\sqrt{2}$

**d.** $\pi/\sqrt{2}$ ✓

$ablaf = (y\cos(xy), x\cos(xy))$, so $ablaf(0, \pi) = (\pi, 0)$; $D_{\hat{u}}f = (\pi, 0) \cdot (1/\sqrt{2}, 1/\sqrt{2}) = \pi/\sqrt{2}$.

**32** [Intermediate] • Target: Lines, exponentials, and trig — the toolkit calculus builds on

Net change theorem: $\int_1^4 v(t)dt$ gives the net displacement if $v(t)$ is velocity. For $v = 2t$:

**a.** $15$ ✓

**b.** $8$

**c.** $30$

**d.** $4$

Applying Fundamental Theorem of Calculus principles, $[t^2]_1^4 = 16 - 1 = 15$. Therefore, the verified correct option is $15$.

**33** [Intermediate] • Target: Factoring and rational expressions

Logistic model: $\frac{dP}{dt} = rP(1 - P/K)$. At $P = K/2$, the growth rate is:

**a.** zero

**b.** minimum

**c.** **maximum** ✓

**d.** negative

Applying Differential Equations principles, Maximum growth at $P = K/2$. Therefore, the verified correct option is maximum.

**34** [Intermediate] • Target: Inverse functions and logarithms

Carrying capacity $K$ in the logistic model: as $P \to K$ from below, growth rate $dP/dt \to$

**a.** $0$ ✓

**b.** $K$

**c.** $\infty$

**d.** $-1$

Applying Differential Equations principles, At $P = K$: $dP/dt = rK(1 - K/K) = 0$. Therefore, the verified correct option is $0$.

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**35** [Intermediate] • Target: Piecewise and absolute value review

Rate of change of volume of sphere $V = \frac{4}{3}\pi r^3$ w.r.t $r$:

**a.** $4\pi r^2$ ✓

**b.** $\frac{4}{3}\pi r^2$

**c.** $\pi r^2$

**d.** $8\pi r$

Applying Applications of Derivatives principles, $dV/dr = 4\pi r^2$. Therefore, the verified correct option is $4\pi r^2$.

**36** [Intermediate] • Target: Systems of equations

Systems: predator-prey (Lotka-Volterra) model has how many equilibria?

**a.** **2** ✓

**b.** 1

**c.** 0

**d.** infinitely many

Applying Differential Equations principles, $(0, 0)$ and $(d/c, a/b)$. Therefore, the verified correct option is 2.

**37** [Intermediate] • Target: Graphing functions calculus will build on

Find the derivative of $f(x) = \operatorname{arsinh}(x)$ with respect to $x$.

**a.** $\frac{1}{\sqrt{x^2+1}}$ ✓

**b.** $\frac{1}{\sqrt{1-x^2}}$

**c.** $\frac{1}{x^2+1}$

**d.** $\frac{1}{\sqrt{x^2-1}}$

By the inverse hyperbolic sine derivative formula, $\frac{d}{dx}[\operatorname{arsinh}(x)] = \frac{1}{\sqrt{x^2+1}}$.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Average rate of change (a preview of the derivative) • Cross-Lesson: 7.01 Integration by Substitution

Related rates: a balloon radius expands at 2 cm/s. Rate of volume change when $r = 5$ cm is:

**a.** $40\pi$ cm³/s

**b.** $100\pi$ cm³/s

**c.** $200\pi$ **cm³/s** ✓

**d.** $300\pi$ cm³/s

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(25)(2) = 200\pi$ cm³/s.

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**39** [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 7.02 Integration by Parts

Frenet equations: $\mathbf{T}' = \kappa\mathbf{N}$, $\mathbf{N}' = -\kappa\mathbf{T} + \tau\mathbf{B}$, $\mathbf{B}' = -\tau\mathbf{N}$. Solving for $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$ given $\kappa(s)$ and $\tau(s)$ determines a curve up to:

**a.** one translation

**b.** **one rigid motion** ✓

**c.** two parameters

**d.** rotation only

Fundamental theorem of space curves: $\kappa$, $\tau$ determine curve up to rigid motion.

**40** [Challenging] • Target: Interval notation and set operations review • Cross-Lesson: 8.01 Area Between Curves

The Net Change Theorem $\int_a^b F'(x)dx = F(b) - F(a)$ requires $F'$ to be:

**a.** continuous and positive

**b.** integrable on $[a, b]$ ✓

**c.** a polynomial

**d.** bounded and increasing

The theorem (a restatement of FTC Part 2) only needs $F'$ integrable on $[a, b]$.

**41** [Challenging] • Target: Functions and composition • Cross-Lesson: 8.02 Volumes of Revolution

Directional derivative $D_{\hat{u}}f = ablaf \cdot \hat{u}$. For $f = x^2y$ at $(1, 1)$ in direction $(1, 1)/\sqrt{2}$:

**a.** $1$

**b.** $\sqrt{2}$

**c.** $3\sqrt{2}$

**d.** $3/\sqrt{2}$ ✓

$ablaf = (2xy, x^2) = (2, 1)$; $D = (2 + 1)/\sqrt{2} = 3/\sqrt{2}$.

**42** [Challenging] • Target: Lines, exponentials, and trig — the toolkit calculus builds on • Cross-Lesson: 8.03 Arc Length and Improper Integrals

The rate of change in the direction $-\hat{u}$ equals:

**a.** $D_{\hat{u}}f$

**b.** $0$

**c.** $|ablaf|$

**d.** $-D_{\hat{u}}f$ ✓

$D_{-\hat{u}}f = ablaf \cdot (-\hat{u}) = -D_{\hat{u}}f$.

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**43** [Challenging] • Target: Factoring and rational expressions • Cross-Lesson: 9.01 Separable Differential Equations

Weierstrass substitution $t = \tan(x/2)$: $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$. Converts rational trig integrands to:

**a.** trig of $t$

**b.** **rational functions of $t$** ✓

**c.** exponentials of $t$

**d.** polynomials

Weierstrass: rationalises all trig expressions.

**44** [Challenging] • Target: Inverse functions and logarithms • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Integration of inverse functions (Laisant-Hermite): $\int_a^b f^{-1}(y)dy = bf^{-1}(b) - af^{-1}(a) - \int_{f^{-1}(a)}^{f^{-1}(b)} f(x)dx$. Example for $f = e^x$:

**a.** gives nothing useful

**b.** yields: $\ln b - \ln a$

**c.** gives $e^b - e^a$

**d.** **yields:** $b\ln b - a\ln a - (b-a)$ ✓

Inverse function integration applied to $\ln$.

**45** [Challenging] • Target: Piecewise and absolute value review • Cross-Lesson: 11.01 Sequences and Series

The directional derivative $D_{\hat{u}}f = ablaf \cdot \hat{u}$. Maximum directional derivative equals:

**a.** $ablaf$

**b.** $|ablaf|$ ✓

**c.** 1

**d.** 0

Max $D_u f = |ablaf|$ (in direction $ablaf$).

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